Dynamic Routing Spatial-Temporal Transformer Model for Traffic Flow Prediction

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Abstract— **Predicting traffic flow is vital component of Intelligent Transportation Systems (ITS), aimed at enhancing urban traffic management and optimization efforts. However, accurate prediction remains a significant challenge due to the vast array of influencing factors. Many existing approaches often overlook the comprehensive impact of diverse factors on prediction accuracy by considering only a subset of data features. To address this critical issue, a Dynamic Routing Spatial-Temporal Transformer (DRSTT) model specifically tailored for traffic flow prediction is proposed. The DRSTT model captures data features through the design of specialized modules that effectively handle short-term variations, long-term trends, static spatial information of road networks, and dynamically evolving spatial patterns. The proposed approach combines dynamic routing techniques with the transformer model to intelligently adjust information transmission paths within the network according to the distinct features of real-time input data. Utilizing its robust self-attention mechanism, the transformer model effectively captures and analyzes the spatial-temporal dependencies present in data, thereby improving the accuracy of traffic flow predictions. Furthermore, distinct position embeddings are incorporated for different modules to further enhance the model's capability to recognize and utilize various feature types. The DRSTT model has been thoroughly tested on three traffic flow datasets, consistently surpassing leading techniques and demonstrating its reliability and efficiency in traffic flow prediction.**

Index Terms— **Transformer, spatial-temporal, dynamic routing, traffic flow prediction.**

I. Introduction

THE transportation system is the lifeblood of modern urban environments, having a significant impact on the daily lives of urban residents and economic development. With the acceleration of urbanization and the increasing number of vehicles on the roads, traffic congestion has become a more severe issue. Accurate traffic flow prediction can provide decision-making support for traffic management, optimize traffic signals, formulate efficient traffic guidance strategies, and alleviate traffic pressure [1].

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The widespread application of big data technologies has provided abundant data sources for traffic flow prediction, such as vehicle detector data, GPS data, and mobile application data, which has led to the growing use of deep learning methods in this field [2], [3]. Traditional methods primarily rely on conventional machine learning techniques within statistical and time series models to capture temporal dependencies [4], [5], [6]. The introduction of Recurrent Neural Networks (RNNs) [7], [8] significantly enhanced the ability to learn these temporal patterns. In recent years, the use of deep learning methods has increased significantly, particularly Graph Neural Networks (GNNs) [9], [10], [11], [12] and Transformer-based models [13], [14], [15]. Additionally, methods based on reinforcement learning for traffic flow prediction have also been explored [16].

Nevertheless, existing models still face significant challenges in handling the dynamics and diversity of traffic patterns, the heterogeneity and uneven complexity of traffic data, especially in addressing sudden events, short-term fluctuations, periodic changes, and regional differences. Current models typically use fixed paths to extract features, making it difficult to adaptively adjust based on changing traffic patterns, resulting in a decline in prediction performance. Liu et al. [17] used a combined method based on deep belief networks for short-term traffic flow prediction. Sun [18] combined temporal convolution with Graph Convolutional Networks (GCNs) to predict long-term traffic flow patterns. Song et al. [11] used multiple modules to extract spatiotemporal correlations from local spatiotemporal graphs at each time interval. Li [19] used data-driven methods to create "temporal graphs" to complement spatial graphs by representing correlations not explicitly indicated. However, these models do not adequately address the spatial and temporal diversity and dynamics.

To address these challenges, this paper proposes a Dynamic Routing Spatial Temporal Transformer (DRSTT) model, which aims to handle complex spatiotemporal dependencies in traffic flow data by introducing a dynamic routing mechanism in combination with the powerful representational capacity of the Transformer architecture. Since traffic flow prediction tasks are highly heterogeneous—different regions' road networks have different spatial features, and traffic flow patterns vary over different time periods—dynamic routing adaptively adjusts the information transmission path, ensuring that the model can selectively focus on key features based on task complexity, rather than using the same computational strategy for all times and locations. This allows the model to effectively capture both long-term and short-term temporal variations, as well as static and dynamic spatial variations during complex situations such as peak periods and accidents, while reducing unnecessary computation during more regular periods. This multi-layered, multi-dimensional feature fusion approach enables DRSTT to exhibit better adaptability and accuracy across various traffic flow prediction tasks. The model fills the gap in the use of dynamic routing in the traffic flow prediction field, breaks the limitations of dynamic routing in image feature extraction, and extends its application to spatiotemporal data analysis. By incorporating the specific task requirements in traffic flow prediction, it enhances prediction accuracy and robustness through the dynamic adjustment mechanism that captures spatiotemporal correlations. To summarize, the key contributions of this paper are:

- An innovative traffic flow prediction framework is proposed, designed to couple and extract multi-scale spatiotemporal features. By creating several specialized modules, the framework is carefully constructed to capture both short-term and long-term temporal variations, while considering the static spatial attributes and dynamically changing spatial patterns of the road network, significantly improving the accuracy of traffic flow predictions;
- For the first time, dynamic routing technology is combined with the Transformer model and applied to traffic flow prediction tasks. This method dynamically adjusts the data flow path to adapt to the real-time characteristics of traffic data. The advanced attention mechanism accurately analyzes time series data, effectively distinguishing and modeling the complex spatiotemporal patterns of traffic flow. Additionally, unique location information embeddings are customized for the feature capturing requirements of each module, optimizing the location index and enhancing the overall performance of the model;
- Extensive experimental results and theoretical analysis consistently demonstrate that the suggested method surpasses existing advanced techniques, emphasizing its superior efficiency in traffic flow prediction.

II. Related Work

A. Traffic Flow Prediction

Predicting urban traffic flow is crucial for the development of smart cities and has garnered substantial academic interest in recent years. Early research in this field primarily relied on traditional time series forecasting methods, such as historical averages [20] and ARIMA models [21]. Although simple and easy to implement, these methods often fail to capture the complex, nonlinear spatial-temporal correlations in traffic data, particularly when handling sudden changes in traffic flow. To address these limitations, machine learning models have gained attention. Techniques such as support vector regression (SVR) [22] and the k-nearest neighbor algorithm [23] were introduced, leveraging either local or global historical data for predictions. However, these methods are generally limited in their ability to handle large-scale data and lack the capacity for deep temporal modeling. The advent of deep learning has significantly advanced the field of traffic flow prediction. Hu et al. [24] proposed using LSTM to mitigate the problem of gradient vanishing in practical applications, but their approach focuses on a single transportation mode and ignores the possible interactions between different transportation modes. Building on this, Cui et al. [25] introduced RNNs with deeply stacked bidirectional and unidirectional LSTM architectures to capture temporal correlations in traffic conditions. However, RNN architectures typically require long sequence inputs, involve high computational costs, and still struggle to capture dynamic changes effectively. Yu et al. [26] employed convolutional neural networks (CNNs) to identify dependencies in Euclidean space. Nevertheless, CNNs exhibit limitations when applied to graph-structured data, such as traffic networks, as they struggle to model complex relationships in non-Euclidean spaces. To overcome these challenges, Li et al. [7] proposed using graph convolutional networks (GCNs) instead of CNNs to better extract spatial dependencies. They introduced the DCRNN model, which treats spatial dependencies in road networks as diffusion processes, thereby modeling non-Euclidean spatial dependencies more effectively. However, this model involves high computational complexity, posing challenges for real-time applications. Building on this, Cui et al. [27] utilized GCNs to capture spatial dependencies and explored learning low-dimensional vertex representations through graph embeddings. Despite these advances, these methods exhibit limited effectiveness in capturing dynamic traffic features and adapting to anomalous traffic events. Attention mechanisms, celebrated for their efficiency and flexibility in modeling dependencies, have found widespread applications across various domains. The integration of graph neural networks (GNNs) with attention mechanisms for spatial-temporal prediction has become a research hotspot. Fang et al. [28] applied attention mechanisms to traffic flow data by combining graph attention networks with trajectory data. They proposed a fine-grained traffic prediction framework that achieved higher prediction accuracy at lower computational costs. However, it still faces challenges in long-term predictions for complex traffic networks.To further explore dynamic graph structures and adaptive modules, Transformer models have been introduced into the traffic prediction domain. These models excel in feature extraction and spatial-temporal relationship modeling, effectively capturing the dynamic changes in traffic flow. However, their efficient implementation requires large-scale data and robust hardware support. A comprehensive discussion of these advancements will be provided in Section II-C.

B. Dynamic Routing

Dynamic routing technology was first introduced by Sabour et al. [29] in the design of Capsule Networks (CapsNet), where it enabled dynamic adjustment of connections between lower-level and upper-level units, facilitating dynamic feature modeling and accurate information transmission. Since then, dynamic routing has been adopted across various domains to address complex tasks. In the field of intelligent transportation systems, Li and Horowitz [30] designed a dynamic routing and queuing management model for mixed autonomy scenarios, where human-driven and autonomous vehicles coexist. By dynamically adjusting path selection and integrating real-time signal responses, this method effectively reduced traffic delays and optimized the efficiency of mixed vehicle traffic, demonstrating the superiority of dynamic routing in path decision-making. Jiang et al. [31] further proposed an energy consumption-based dynamic routing algorithm (DR-EC), which combines multi-step prediction and dynamic updating mechanisms to optimize energy consumption and time efficiency for electric vehicle route planning. This algorithm dynamically adjusts road weights, avoiding frequent and inefficient re-planning. In other domains, dynamic routing has also seen widespread application. Wang et al. [32] proposed a dynamic fuzzy routing algorithm (DFRDRL) for software-defined networking (SDN), integrating deep reinforcement learning (DRL) and dynamic traffic prediction, thereby significantly improving network traffic management efficiency. Zhou et al. [33] developed a dynamic routing-based model for multimodal tasks such as visual question answering (VQA) and referring expression comprehension (REC). By dynamically adjusting the level of visual information from macro to micro, this model optimized attention mechanisms and reasoning processes, showcasing the applicability of dynamic routing in complex task decomposition. Despite the flexibility and modeling capabilities demonstrated by dynamic routing technology across various fields, existing research still has limitations. Dynamic routing has predominantly focused on path planning and vehicle management, with little to no application in the critical task of traffic flow prediction. Furthermore, existing methods mainly optimize path selection or signal responses, lacking systematic handling of spatiotemporal dynamic features, which makes them less capable of addressing complex, high-dimensional, and dynamic traffic scenarios. To address these limitations, this paper proposes a novel spatiotemporal prediction model based on dynamic routing, aiming to fill this research gap.

C. Transformer

Since the Transformer model was proposed by Vaswani et al. [34] in 2017, it has developed into a new deep learning architecture that utilizes the attention mechanism and positional encoding strategy for sequence modeling. Li and Moura [35] were the first to apply the Transformer architecture to traffic flow prediction, using graph structure learning to represent the spatial dependencies between data points in different locations. Based on the graph's topological structure, they sparsified the Transformer. However, the model's strong dependence on the graph structure makes it difficult to adapt to the dynamic changes in traffic networks. To further enhance the capture of spatiotemporal features, Xu et al. [14] proposed a Spatial-Temporal Transformer Network (STTN). The model uses a multi-head attention mechanism to jointly model various spatial dependency patterns, considering factors such as similarity, connectivity, and covariance. However, the model does not comprehensively examine the dynamic nature of spatial features, making it difficult to fully capture the dynamic changes in spatial relationships. Meanwhile, Cai et al. [36] focused on capturing ongoing patterns and periodic features of time series and modeling the complex spatial dependencies of traffic flow. By combining global encoders and local-global decoders, they successfully extracted and integrated multiple features to explain the complex spatial relationships in transportation networks. However, the paper only considers temporal attention mechanisms and does not address spatial features at a more refined scale.Zhang et al. [37] captured short-term and long-term temporal dependencies through a multi-head attention mechanism. They also used a fusion decoder to incorporate various types of input, which improves prediction accuracy. However, the paper only considers temporal features and overlooks the impact of spatial dependencies on prediction accuracy. The Bi-STAT model [38] introduces spatially adaptive and temporally adaptive Transformer structures within an encoder-decoder framework. Through a dynamic halting module, the model dynamically allocates computational resources based on task complexity, optimizing performance for different prediction challenges. However, separating the recollection task from the prediction task may result in insufficient integration of the two tasks, potentially affecting overall prediction accuracy. The impact of the Transformer model has also extended to fields such as transportation demand forecasting and traffic pattern recognition, demonstrating its versatility and ability to handle complex, multidimensional data across various applications.

III. Methods

A. Problem Formulation

Traffic flow prediction is a process that involves utilizing historical traffic data as a basis for predicting future traffic conditions. Essentially, this endeavor can be envisioned as a mapping function, which converts gathered traffic data into anticipated traffic states, including metrics such as future traffic flows and speeds. The fundamental objective of this prediction lies in establishing a reliable and robust linkage between historical and future data.

In this model, the focus is on a study area equipped with N sensors, each collecting traffic flow data in a fixed time interval Δt. Starting from the current time t, the prediction period extends to P. Traffic flow data throughout this entire period can be expressed by a three-dimensional tensor $X_G^{t+1:t+P}$ with dimensions $\mathbb{R}^{N\times P\times d}$, where d represents the dimension of the traffic state (in this study, we consider a single state, traffic flow, thus $d = 1$). The traffic flow collected during the i^{th} time step from the current time point can be denoted as $X_G^i \in \mathbb{R}^{N\times d}$. A traffic network can be represented by a graph $G(V, E_{\text{roads}}, A)$, where $V = \{v_1, v_2, \ldots, v_N\}$ denotes the set of vertices, corresponding to the locations of the sensors in the network; E_{road} represents the set of edges, indicating the connections between sensors; and $A \in \mathbb{R}^{N\times N}$ denotes the adjacency matrix, which quantifies the strength of these connections. Based on traffic flow data $X_G^{t-T+1:t}$ at historical time step T, the traffic flow prediction problem involves finding a function f that forecasts traffic flow for the next Q steps, starting from the current time t, denoted as $t+1 : t+Q$, where Q is the number of future time steps to be predicted. Mathematically, This can be expressed through the following process:

$$\hat{Y}_G^{t+1:t+Q} = f(X_G^{t-T+1:t}) \tag{1}$$

B. Framework Overview

The DRSTT model is meticulously crafted to achieve precise and reliable traffic flow predictions. As shown in Fig. 1, the suggested DRSTT model consists of three main components: the input layer, the DRSTT module, and the output layer. The input sources integrate traffic flow data, time embeddings, and a spatial adjacency matrix. Within the DRSTT framework, two custom-designed transformer modules are utilized to extract temporal evolution patterns. The Dynamic Routing Long Temporal Transformer (DynRLTT) targets long-term temporal features, whereas the Dynamic Routing Short Temporal Transformer (DynRSTT) addresses short-term temporal characteristics. These temporal features are then combined through a fusion layer called the Temporal Gated Fusion Layer (TGFL), creating a unified temporal profile. The framework includes two additional modules to handle spatial features. The Dynamic Routing Dynamic Spatial Transformer (DynRDST) captures the dynamic spatial features present in the data, while the Dynamic Routing Static Spatial Transformer (DynRSST) captures the detailed aspects of the static structure of the traffic road network. These spatial features are merged by the Spatial Gated Fusion Layer (SGFL) to provide a thorough understanding of spatial characteristics. The DRSTT module employs various location coding strategies to represent the spatial locations of different features, enhancing the model's sensitivity to spatial-temporal details. This detailed integration of temporal and spatial features creates a comprehensive feature set with rich spatial-temporal information, which is then processed by the output layer to accurately predict future traffic flows. Through this sophisticated design, the DRSTT model adeptly manages complex spatial-temporal relationships and exhibits remarkable adaptability to dynamic changes in traffic conditions. This feature highlights its effectiveness and dependability in managing and organizing traffic, offering accurate and practical information to enhance predictions for traffic flow.

C. Preprocessing

1) Time Embedding: To identify temporal patterns in traffic flow data and improve prediction accuracy, the model integrates detailed temporal information. Since there are significant differences in traffic flow between weekdays and weekends, as well as between peak and non-peak hours within a day, we consider two important temporal signals: the day of the week and the time step of the day.

Specifically, a week is divided into 7 days, and a 7-bit one-hot vector is used to represent the day of the week when the data is collected. For instance, if the first bit is 1, the current vector represents Monday, and so on. This 7-bit one-hot encoding effectively captures the information of which day of the week it is. Similarly, a day is divided into M time steps, and an M-bit one-hot vector is used to represent the specific time step within the day. These two types of temporal information are concatenated into a single feature vector with a dimension of R^{7+M}, which fully expresses the temporal characteristics of the current time.

This feature vector contains rich temporal information and is subsequently passed through two fully connected layers to further extract temporal patterns and trends. The final output vector serves as the temporal embedding TE of the model. This temporal embedding method effectively helps the model capture time-related patterns and trends, thereby improving the prediction accuracy of traffic flow. The specific methodology is illustrated in Fig. 2.

2) Adjacency Matrix: The initial step involves constructing a spatial adjacency matrix that accurately represents the physical connections between various road sections based on the actual road layout. In this matrix, the initial edge weights between nodes (which denote road segments) embody the non-Euclidean topological relationships among them. To further refine the spatial adjacency matrix, a distance metric ρ based on the actual traffic road network and a threshold standard deviation k are introduced. These parameters direct the use of a threshold Gaussian kernel function, which is applied to adjust and reconstruct the spatial adjacency matrix A. Through this process, the weights are recalibrated to reflect the physical distances between nodes more precisely, thereby enhancing the matrix's representation of actual spatial relationships. The specific calculations are as follows:

$$A_{ij} = \begin{cases} \exp\left(-\dfrac{d_{ij}^2}{\rho^2}\right), & d_{ij} \le k \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where A_{ij} is the edge weight between node v_i and node v_j. Let d_{ij} denote the actual road network distance between nodes v_i and v_j, ρ represent the standard deviation of the actual road network distance, and k be the set threshold.

D. Dynamic Routing Temporal Transformer

This study introduces the DynRLTT module, meticulously designed to capture and analyze long-term temporal dependencies in traffic flow. Recognizing the critical importance of short-term fluctuations for rapid adaptation to unforeseen events, the study also incorporates the DynRSTT module specifically designed to extract and interpret short-term temporal attributes. This dual strategy significantly enhances the model's ability to fully grasp and forecast the intricate patterns of traffic flows. Traffic flow data, temporal embedding and adjacency matrix were processed as model inputs through a convolutional layer as shown in Fig. 1.

1) Dynamic Routing Long Temporal Transformer: To effectively capture long-term temporal features in traffic flow, this study introduces periodic encoding. This approach leverages an inverse time scale to modulate the frequency of features across various dimensions, enabling the model to detect variations at multiple temporal scales. The *embeddings* are generated using sine and cosine functions applied on a

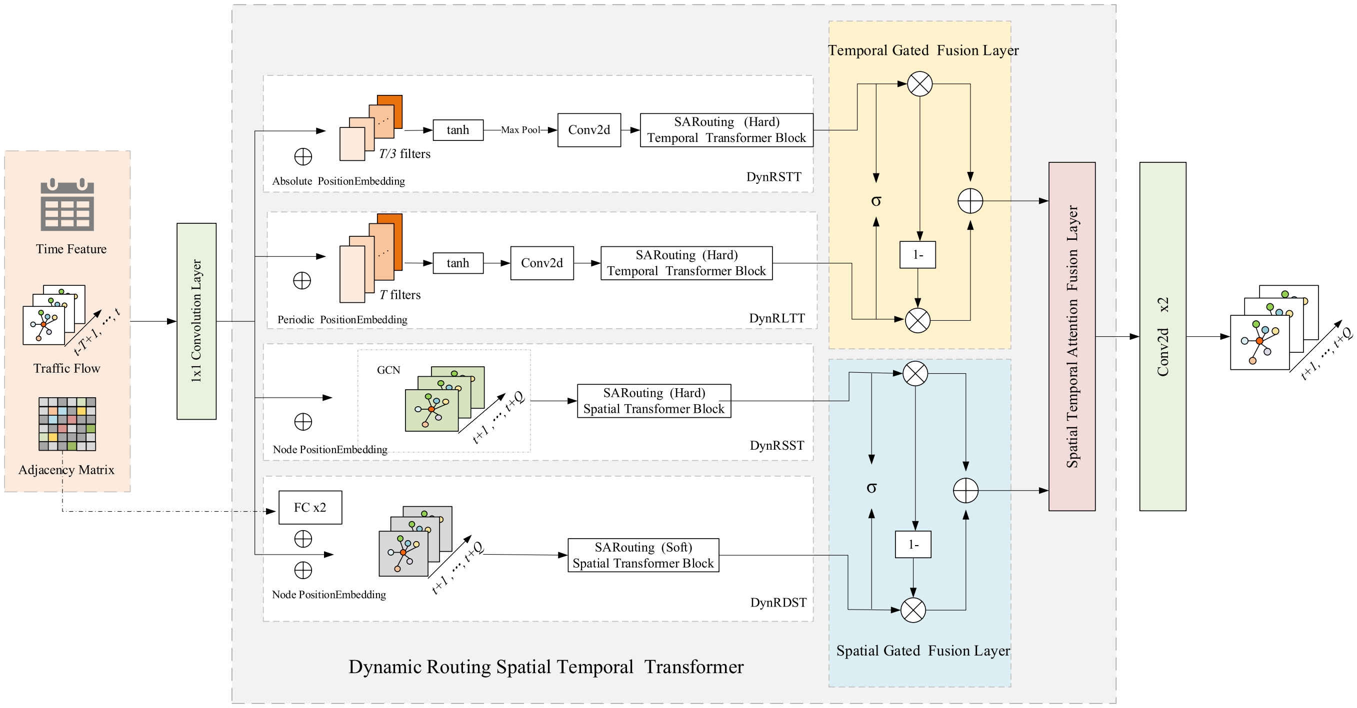

Fig. 1. The framework of the DRSTT.

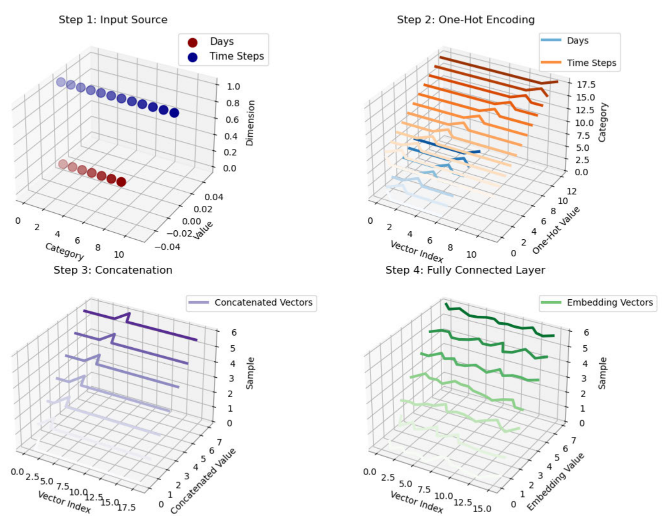

Fig. 2. A flowchart about time embedding.

logarithmic time scale. This positional encoding technique adeptly captures periodic features at different frequencies, which is essential for comprehending and forecasting the recurrent patterns inherent in traffic flow. The mathematical process is detailed below:

$$PE_{\text{periodic}}(P_T, 2i + j) = \sin\left(\frac{P_T \times q_j}{10000^{2i/D}}\right) \tag{3}$$

$$PE_{\text{periodic}}(P_T, 2i + 1 + j) = \cos\left(\frac{P_T \times q_j}{10000^{2i/D}}\right) \tag{4}$$

where P_T is the location index of the time step, q_j is the j inverse time scale, and i is the dimension of the embedding. As a larger receptive field helps in obtaining global temporal information, the DynRLTT module increases the perceptual field of the time view. In this module, a global receptive field is achieved by covering the entire input sequence using a T-filter of size $(T, 1)$. The input data, utilizing periodic position encoding for sequence position indexing, enables the model to understand the long-term relationships between different time points in the time series. The resulting output is as follows:

$$X_{LT} = \phi\,(c_1 X + a) \tag{5}$$

where ϕ is the tanh activation function, c_1 represents the convolution operation, and a is the learnable bias parameter. The resulting outputs are subsequently input into the Self Attention Routing (SARouting) (Hard) Temporal Transformer Block module, as illustrated in Fig. 3, which dynamically adjusts its structure based on the input data. Despite some fluctuations, long-term temporal features maintain a fundamental regularity and stability over extended periods. Hard routing is particularly effective in capturing these enduring patterns through a static path structure, which facilitates stable feature transmission and robust modeling. This approach empowers the model to identify and exploit long-term trends within the data. In this process, routing weights are determined, then weighted and aggregated across the outputs of the self-attention mechanism. This integration yields the final routing output, as detailed in Section III-F.

$$\bar{X}_{LT} = Routing\,(X_{LT}) \tag{6}$$

The refined output is then integrated into the temporal multi-head self-attention mechanism, further improving the DRSTT model's capability to process and understand long-term temporal dependencies. For each attention head, the attention weights are computed using query Q, key K, and value V.

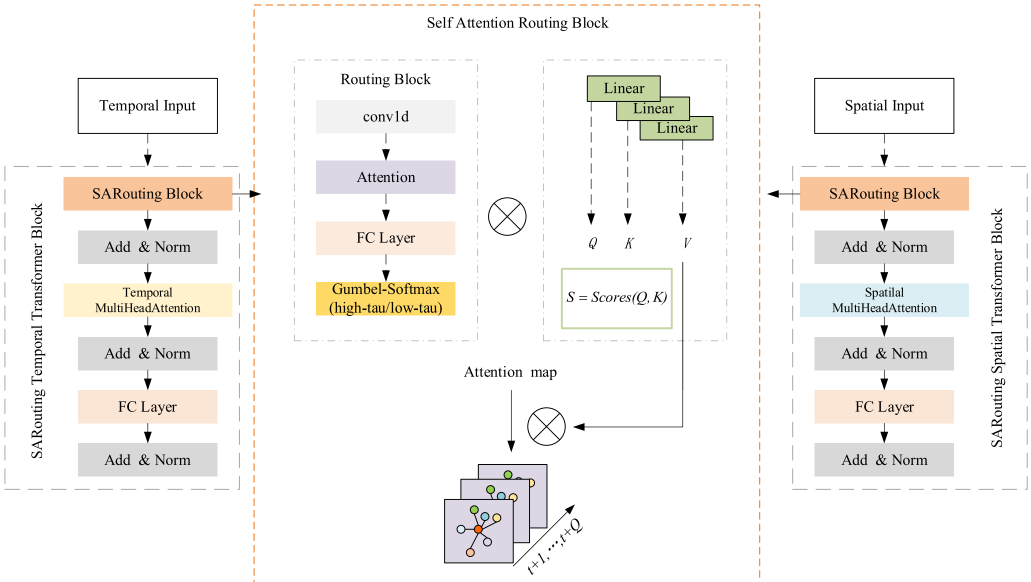

Fig. 3. The framework of the SARouting Spatial(Temporal) Transformer Block.

The weighted values are then calculated. Here, W_{q_i}, W_{k_i}, W_{v_i} is the learnable weight matrix. In the multi-head self-attention model, we receive three input matrices: the query matrix X_q, the key matrix X_k, and the value matrix X_v. By applying linear transformations, we generate corresponding query subspace Q_i, key subspace K_i, and value subspace V_i for each input matrix, which are located in $R^{d_m \times d_q}$, $R^{d_m \times d_k}$, and $R^{d_m \times d_v}$ spaces.

$$\begin{aligned} Q_i &= X_q W_{q_i}, \\ K_i &= X_k W_{k_i}, \\ V_i &= X_v W_{v_i}, \end{aligned} \tag{7}$$

Use these subspaces to compute attention scores:

$$Score = \frac{Q_i K_i^T}{\sqrt{d_k}} \tag{8}$$

where d_k is the scaling factor. We utilize the attention score to calculate the attention result of the i^{th} head:

$$Head_i = Att(Q_i, K_i, V_i) = \text{softmax}(Score)V_i \tag{9}$$

Subsequently, a multi-head self-attention unit is employed to concatenate the features from all h attention heads. This unit facilitates the learning of the final attention weights through a feed-forward neural network, enabling further feature extraction and nonlinear transformation. The output $MH(X_q, X_k, X_v)$ can be expressed as:

$$MH(X_q, X_k, X_v) = \text{cat}(Head_1, \ldots, Head_h)W_o \tag{10}$$

where cat denotes the concatenation of the outputs of all heads, while W_o is a trainable weight matrix that integrates the attention features of different heads and performs dimensionality reduction. For this module, which processes long-term temporal features, we construct a temporal multi-head attention layer to process the extracted long-term time series features $\hat{X}_{LT} \in \mathbb{R}^{T \times F}$, and F denote the embedding dimension. These temporal feature sequences are embedded into a higher-dimensional space, allowing the model to deeply analyze the intricate and abstract temporal relationships within the time series data. This embedding process facilitates a deeper understanding of the complex temporal relationships that exist within the dataset. Through the application of the multi-head attention mechanism, the model further refines and interprets these dependencies, it can be derived:

$$\hat{X}_{LT} = MH(\bar{X}_{LT}, \bar{X}_{LT}, \bar{X}_{LT}) \tag{11}$$

The long-term temporal attention features thus extracted are then fed into subsequent layers of the model for further processing.

2) Dynamic Routing Short Temporal Transformer: To enhance the understanding of the chronological sequence of time points, the DynRSTT module employs absolute positional encoding. This technique explicitly delineates the positional hierarchy of each data point within the temporal sequence by generating a unique encoding for each individual time point. By leveraging a sine function to assess the positional relationships among the various elements within the sequence, the model emphasizes interactions based on their relative proximity rather than solely considering their global positions within the sequence. This approach effectively captures the intricate relationships between adjacent elements. Indicates the following:

$$PE_{\text{absolute}}(P_T, i) = \sin\left(\frac{P_T}{10000^{2i/D}}\right) \tag{12}$$

where P_T is the location index of the time step and i is the dimension of the embedding. To reduce the perceptual domain in the time view, a $(T/3, 1)$ filter of size T is used. Given input $X \in \mathbb{R}^{H \times T \times N}$, represented as follows:

$$X_{ST} = \phi\,(c_2 X + b) \tag{13}$$

where ϕ is the tanh activation function, c_2 is the convolution operation, and b is the learnable bias parameter. A window approximately two-thirds the size of the temporal dimension is employed to select the maximum value within each window along the temporal axis, while preserving the feature dimension intact. This pooling operation efficiently minimizes the temporal span of the time series data, enabling the model to identify essential features within the sequence while maintaining the integrity of the feature dimension. The output of this operation is then fed back into the SARouting (Hard) Temporal Transformer Block, as illustrated in Fig. 3. Given the inherently higher predictability and stability of short-term temporal features, hard routing is particularly well-suited for processing and transmitting these features through predefined pathways. This approach enhances computational efficiency and real-time performance. Due to the limited variability of short-term features, the rigid structure of hard routing is sufficient to manage the fluctuations effectively.

By concentrating on extracting short-term temporal features from traffic flow data, the SARoutingBlock (Hard) generates the final routing output, as explained in Section III-F.

$$\bar{X}_{ST} = Routing\,(X_{ST}) \tag{14}$$

This output is then incorporated into the temporal multi-head self-attention mechanism, as specified in (9). A temporal multi-head attention layer is created to process the extracted short-term time series features, denoted as $\hat{X}_{ST} \in \mathbb{R}^{T\times F}$, and F represents the embedding dimension. These temporal feature sequences are embedded into a higher-dimensional space, allowing the model to learn deeper and more abstract temporal dependencies within the time series. This attention mechanism facilitates the comprehensive analysis of temporal relationships by capturing nuanced interactions within the sequence. Denotes as:

$$\hat{X}_{ST} = MH(\bar{X}_{ST}, \bar{X}_{ST}, \bar{X}_{ST}) \tag{15}$$

The short-term temporal attention features thus extracted are then fed into subsequent layers of the model for further processing.

E. Dynamic Routing Spatial Transformer

Traffic flow conditions are shaped by a diverse array of temporal factors and spatial elements. To address these complex challenges, this paper introduces the DynSST module, which adeptly extracts static spatial features, providing a deep understanding of the fundamental road layout. Additionally, the DynDST module is designed to capture dynamic spatial features, reflecting the status and changing trends of traffic flow within the road network. This dual approach significantly enhances the precision of traffic flow predictions. By meticulously analyzing static features, structural issues within the traffic system can be identified and addressed. Simultaneously, the continuous monitoring and responsive management of dynamic features enable effective handling of unexpected events and temporary fluctuations in traffic flow.

1) Dynamic Routing Static Spatial Transforme: To precisely capture spatial characteristics within the traffic network, a new positional encoding scheme has been devised, differing from previously discussed methods. This approach uses sine and cosine functions applied on a logarithmic timeline to create positional indices for individual nodes. These indices serve as positional markers within the sequence, allowing the model to thoroughly understand and predict traffic flow patterns across different nodes and time intervals. This innovative encoding method effectively captures the complex interactions among various nodes in the traffic network. By offering a precise spatial context for each node, the model is better able to analyze the complex behavior of traffic flow. Details of the calculation are below:

$$PE_{\text{nodes}}(P_N, 2i) = \sin\left(\frac{P_N}{10000^{2i/D}}\right) \tag{16}$$

$$PE_{\text{nodes}}(P_N, 2i+1) = \cos\left(\frac{P_N}{10000^{2i/D}}\right) \tag{17}$$

where P_N is the position index of the node and i is the dimension of the embedding. This module processes raw traffic flow data and an adjacency matrix as inputs, incorporating sequence positional information through node position encoding. When handling graph-based data, a specialized convolutional framework is utilized to rapidly propagate node states across the network structure. By constructing filters in the frequency domain, this framework adeptly captures the spatial correlations among nodes and their neighboring nodes within the traffic network. The first step involves computing the degree matrix D of the adjacency matrix:

$$d_i = \sum_{j=1}^{N} A_{ij} \tag{18}$$

$$D = \text{diag}(d_1^{-\frac{1}{2}}, d_2^{-\frac{1}{2}}, \ldots, d_N^{-\frac{1}{2}}) \tag{19}$$

For the traffic flow feature matrix $X \in \mathbb{R}^{N\times T\times W}$, H represents the width of the feature, T represents the time step, and N represents the number of nodes. The spatial information is propagated and aggregated through a series of convolution operations. Each convolution operation can be represented as:

$$F_{\text{conv}} = \sigma(D \cdot A \cdot D \cdot X \cdot W_a) \tag{20}$$

where σ is the ReLU activation function, W_a is a trainable weight matrix. With the above convolution, the corresponding output is obtained:

$$X_{SS} = F_{\text{conv}}\,(X, A) \tag{21}$$

The resultant output is subsequently channeled into the SARouting (Hard) Spatial Transformer Block, as illustrated in Fig. 3. Given that static spatial attributes, such as road networks and traffic infrastructure, are constant and unchanging, the hard routing approach is particularly effective in encoding and transmitting these characteristics. This method ensures the stable transmission of paths and minimizes unnecessary computational resource expenditure. Since these features remain invariant over time, hard routing maintains their stability and consistency, allowing the model to leverage these fixed attributes with greater precision. Consequently, the model can focus on extracting static spatial features from the data more effectively. The ultimate routing output is derived from the

SARoutingBlock (Hard), with detailed specifics provided in Section III-F.

$$\bar{X}_{SS} = Routing\,(X_{SS}) \tag{22}$$

The refined output is then seamlessly incorporated into the spatial multi-head self-attention mechanism. As detailed in (9), a spatial multi-head attention layer is built to process the extracted static spatial features $\hat{X}_{SS} \in \mathbb{R}^{N\times F}$, and F represents the embedding dimension. Through this attention mechanism, the computation process is as follows:

$$\hat{X}_{SS} = MH(\bar{X}_{SS}, \bar{X}_{SS}, \bar{X}_{SS}) \tag{23}$$

The static spatial features thus extracted are then passed into subsequent layers of the model for further processing.

2) Dynamic Routing Dynamic Spatial Transformer: This module incorporates node positional encoding to provide positional indices, following the methodology outlined in the DynSTT module. To effectively identify and extract potential spatial dependencies in traffic flow data and adjacency matrices, the model continuously adapts to changing conditions and updates its understanding of the graph structure. This is achieved by using a sequence of densely interconnected neural network layers to convert the static adjacency matrix into a series of dynamic adjacency matrices. At each temporal interval, these network layers generate a new adjacency matrix that precisely reflects the current traffic network configuration and spatial interconnections. The updated adjacency matrix at time step is calculated as follows:

$$A^t_{FC} = \phi(\sigma(A \cdot W_y)W_r) \tag{24}$$

where ϕ, σ are ReLU activation functions, and W_y, W_r are trainable weight matrices. Through the above process, the corresponding output is obtained:

$$X_{DS} = A^t_{FC}\,(X, A) \tag{25}$$

The resulting output is then fed into the SARouting (Hard) Spatial Transformer Block, as shown in Fig. 3. Dynamic spatial features, such as is real-time traffic conditions and the impacts of incidents, are characterized by frequent fluctuations and inherent uncertainty. Soft routing, with its capacity for flexible path adjustment, assigns varying attention weights to information from different spatial locations. This flexibility enables the model to dynamically adjust its emphasis on spatial features based on real-time inputs, thereby improving its capacity to capture real-time variations in dynamic systems like traffic flow. The SARoutingBlock (Soft) is employed to generate the ultimate routing output, which is further detailed in Section III-F.

$$\bar{X}_{DS} = Routing\,(X_{DS}) \tag{26}$$

This output is subsequently integrated into the spatial multi-head attention mechanism, facilitating a comprehensive analysis of the spatial features. As described in (9), a spatial multi-head attention layer is constructed to process the extracted dynamic spatial features $\hat{X}_{DS} \in \mathbb{R}^{N\times F}$, and F denote the embedding dimension. Through this attention mechanism, the computation process is as follows:

$$\hat{X}_{DS} = MH(\bar{X}_{DS}, \bar{X}_{DS}, \bar{X}_{DS}) \tag{27}$$

The dynamic spatial features thus extracted are then passed into subsequent layers of the model for further processing.

F. Self Attention Routing Block

In the realm of traffic flow prediction, data characteristics exhibit significant variations across different road segments and temporal periods. Throughout the entire process, considerable uncertainty persists, arising from fluctuations in traffic conditions, data noise, and a myriad of other influencing factors. To effectively address this uncertainty, a dynamic routing mechanism is employed, enabling the adaptive adjustment of information transmission paths based on the inherent uncertainty of the input data. This mechanism not only enhances the model's capability to manage uncertainty but also optimizes the utilization of information, thereby mitigating the adverse effects of uncertainty on prediction outcomes. This paper introduces the SARoutingBlock module, a pivotal component that integrates temporal and spatial features derived from the preceding modules. Within this module, convolutional layers are employed to capture the inherent patterns and salient features within the sequences. Subsequently, routing weights are meticulously calculated based on the intricate relationships present in the input data. By employing this comprehensive approach, the model can dynamically adapt to varying traffic conditions, thereby greatly improving the precision and dependability of traffic flow predictions.

$$RAtt(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)VW_m \tag{28}$$

where Q, K, V are results obtained from (7), W_m is the learnable weight matrix. These weights determine the information transmission and routing process between each time step and are subsequently input into the fully connected layers. However, the resulting features may be highly complex and diverse. By using Gumbel-Softmax, the weights of different features can be refined, making the model more stable and robust in feature selection. The final routing weight δ is calculated as follows:

$$g_i = -\log(-\log(U_i)), \quad U_i \sim U(0, 1) \tag{29}$$

$$\delta = \frac{\exp((\log(X_i) + g_i)/\tau)}{\sum_{j=1}^{k}\exp((\log(X_j) + g_j)/\tau)} \tag{30}$$

where g_i represents the Gumbel noise, τ is the temperature parameter, and k is the number of features. Specifically, the temperature parameter τ regulates the smoothness of the distribution. When the temperature τ is low, the resulting distribution is sharp, making the module closer to hard routing. Conversely, when the temperature τ is high, the resulting distribution is smoother, making the module closer to soft routing. The input to this module is subsequently passed through a linear layer to compute the attention scores, as demonstrated in (8):

$$S = \text{Scores}(Q_r, K_r) \tag{31}$$

where Q_r, K_r, V_r is the result of the calculation in (7). The attention output is weighted and aggregated to produce the final routing output:

$$\text{Routing} = S \cdot \delta \cdot V_r \tag{32}$$

The subsequently acquired output is subsequently introduced into the model's temporal transformer or spatial transformer for further elaboration, thereby affording a deeper understanding of the internal sequence structure.The algorithm table is given below to elaborate on the working of SARouting Spatial (Temporal) Transformer Block, using long term temporal features as input.

Algorithm 1 SARouting

Require: Long-term time features $\mathbf{X}_{\text{LT}}$

1: Pass the input $\mathbf{X}_{\text{LT}}$ through a Conv1D layer to extract features, resulting in feature matrix $\mathbf{X}'$
2: Compute the Q, K, V matrices (linear transformation), using Eq. (7)
3: Compute self-attention via $RAtt(Q, K, V)$, using Eq. (28)
4: Use Gumbel-Softmax to adjust routing weights δ, using Eq. (29) and Eq. (30)
5: Compute the final weighted routing output $\bar{\mathbf{X}}_{\text{LT}} \leftarrow (\mathbf{X}_{\text{LT}})$, using Eq. (6)
6: Pass the attention output to a fully connected layer or subsequent Transformer module $\hat{\mathbf{X}}_{\text{LT}} \leftarrow (\bar{\mathbf{X}}_{\text{LT}})$, using Eq. (11)
7: **Return** Long-term time features $\hat{\mathbf{X}}_{\text{LT}}$

G. Gated Fusion Layer and Temporal-Spatial Transformer

1) Temporal Gated Fusion Layer: Using the extracted long-term and short-term time features, this paper designs a module to fuse them. Let $\hat{X}_{LT}$ represent the long-term time feature and $\hat{X}_{ST}$ represent the short-term time feature. Initially, a linear model with Sigmoid activation is employed to obtain the gating values, as follows:

$$g_T = \theta(\hat{X}_{LT} W_l + \hat{X}_{ST} W_s) \tag{33}$$

where W_l, W_s are the learnable weight matrices, and $\theta(\cdot)$ is the activation function of Sigmoid. The gating values g_T are then combined to weight the fusion of the two input features, calculated as follows:

$$\hat{X}_T = g_T \cdot \hat{X}_{LT} + (1 - g_T) \cdot \hat{X}_{ST} \tag{34}$$

2) Spatial Gated Fusion Layer: This module is consistent with the computation of the time-gated attention fusion layer. Let $\hat{X}_{SS}$ and $\hat{X}_{DS}$ denote the static and dynamic spatial features. The spatial fusion feature $\hat{X}_S$ is calculated from (28):

$$\hat{X}_S = g_S \cdot \hat{X}_{SS} + (1 - g_S) \cdot \hat{X}_{DS} \tag{35}$$

3) Temporal-Spatial Transformer: Employing the multi-attention mechanism, the temporal features $\hat{X}_T$ and spatial features $\hat{X}_S$ acquired earlier are fused to compute the ultimate spatial-temporal features, depicted in (9):

$$\hat{X} = MH(\hat{X}_S, \hat{X}_T, \hat{X}_T) \tag{36}$$

TABLE I

DATASET STATISTICS

Datasets	Nodes	Time steps	Time duration
PEMS03	358	26208	09/2018 – 11/2018
PEMS04	307	16992	01/2018 – 02/2018
PEMS08	170	17856	07/2016 – 08/2016

H. Output Layer and Loss Function

1) Output Layer: The fused spatial-temporal features mentioned above are inputted into a prediction layer comprising two 1×1 convolutional layers, denoted as:

$$\hat{Y} = c_3(\delta(c_4 \hat{X} + d)) + e \tag{37}$$

where c_3 and c_4 are two independent convolution operations, d and e are learnable parameters, and $\delta(\cdot)$ is the activation function of ReLu.

2) Loss Function: Since SmoothL1Loss has better numerical stability when the error is small and stronger robustness when the error is large. During training, the following loss function is used to minimize the error between the true value Y and the predicted value $\hat{Y}$:

$$Loss(\hat{y}_i, y_i) = \begin{cases} \frac{1}{2}(\hat{y}_i - y_i)^2 & \text{, if } |\hat{y}_i - y_i| < 1 \\ |\hat{y}_i - y_i| - \frac{1}{2} & \text{, otherwise} \end{cases} \tag{38}$$

where $\hat{y}_i, y_i$ describe the predicted and actual traffic flow.

IV. EXPERIMENTS

This section concentrates on performing a set of experiments of the proposed framework. Initially, the datasets utilized for these experiments are introduced, along with the performance evaluation metrics. Subsequently, the experimental configuration is described, and the findings are presented alongside a detailed discussion. Additionally, ablation studies and visualizations are performed to provide deeper insights into the framework's functionality and to further elucidate its operational dynamics.

A. Datasets and Evaluation Metrics

1) Dataset: The evaluation involved three datasets specifically for traffic flow prediction: PEMS03, PEMS04, and PEMS08. Originating from various regions within California, these datasets are part of the Performance Measurement System (PeMS) managed by the California Department of Transportation. PeMS provides real-time monitoring of freeway traffic conditions throughout the state [39]. The prediction tasks were formulated based on the traffic flow characteristics inherent in these datasets. Comprehensive details about the datasets are provided in TABLE I.

The raw data is gathered in real-time at 30-second intervals. However, the benchmark datasets used in this study present aggregated data, compiled at 5-minute intervals, resulting in a total of 288 time points per day.

2) Evaluation Metrics: In the experiments, the model's precision is measured using three commonly employed metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

TABLE II

COMPARISON OF THE EXPERIMENTAL RESULTS OF THE MODEL IN THIS PAPER WITH TEN BENCHMARK MODELS IN THE DATASETS PEMS03, PEMS04, PEMS08

Model	Metric	DCRNN	AGCRN	Graph WaveNet	STGCN	STSGCN	STFGNN	GMAN	STTN	Traffic Transformer	Bi-STAT	COGCN	GHTPAN	DRSTT
PEMS03	MAE	18.18	16.75	19.85	17.49	17.48	16.77	16.49	16.11	16.39	15.29	15.72	15.50	**15.14**
	RMSE	30.31	28.60	32.94	30.12	29.21	28.34	26.48	27.87	27.87	27.54	27.34	**24.38**	26.79
	MAPE(%)	18.91	16.23	19.31	17.15	16.78	16.30	17.13	16.19	15.84	15.19	15.27	15.92	**15.10**
PEMS04	MAE	24.70	19.83	25.45	22.70	21.19	19.83	19.36	19.32	19.16	18.74	19.63	20.84	**18.31**
	RMSE	38.12	32.26	39.7	35.55	33.65	31.88	31.06	30.79	30.57	30.31	31.68	30.87	**30.08**
	MAPE(%)	17.12	12.97	17.29	14.59	13.90	13.02	13.55	13.15	13.70	12.59	13.26	13.80	**12.07**
PEMS08	MAE	17.86	15.95	19.13	18.02	17.13	16.64	14.51	15.28	15.37	14.07	15.63	16.82	**13.48**
	RMSE	27.83	25.22	31.05	27.83	26.80	26.22	23.68	24.25	24.21	23.45	24.28	24.90	**22.79**
	MAPE(%)	11.45	10.09	12.68	11.40	10.96	10.60	9.45	9.98	10.09	9.27	10.85	11.00	**9.03**

B. Baselines

To validate the effectiveness of our approach, DRSTT was compared with the following 12 benchmark models:

- DCRNN [7]: Depicts traffic flow as a diffusion process on directed graphs, using an encoder-decoder framework to capture spatial and temporal relationships.
- AGCRN [40]: Employs adaptive modules to autonomously identify spatial and temporal correlations in traffic sequences, removing the requirement for predefined graphs.
- STGCN [10]: Employs graph convolutional frameworks to efficiently capture multi-scale spatial-temporal correlations within traffic networks.
- Graph WaveNet [41]: Leverages adaptive dependency matrices and dilated convolutional components to accurately capture spatial dependencies and handle long-sequence temporal data.
- STSGCN [11]:Adopts a spatial-temporal synchronous modeling mechanism to identify intricate local spatial-temporal relationships.
- STFGNN [19]: Creates time graphs and combines multiple spatial and temporal graphs from different periods to efficiently learn spatial-temporal dependencies.
- GMAN [42]: Reduces error propagation in long-term predictions using spatial and temporal attention mechanisms.
- Traffic Transformer [35]: Designs temporal sequence encoding strategies and incorporates graph convolutional networks to grasp complex spatial-temporal correlations in traffic data.
- STTN [14]: Combines dynamic spatial dependencies and long-term temporal dependencies through spatial and temporal transformers to grasp dynamic spatial-temporal correlations in traffic flow.
- Bi-STAT [38]: Dynamically handles the intricacies of spatial and temporal tasks and enhances future traffic prediction accuracy by incorporating past traffic conditions.
- COGCN [43]: The model optimizes graph convolutional networks via contrastive learning, linking optimized road network graphs while preserving global-local feature consistency. It performs well on speed and traffic flow datasets.
- GHTPAN [44]: A head-tail partner attention module is introduced. By employing a dual mechanism that includes channel attention and weighted assignment, the system intelligently recognizes sensor nodes in the road network and assigns appropriate weights to them.

C. Experiment Setting

The experiments were performed on a server featuring an NVIDIA Tesla T4 GPU and 48 GB of memory, using PyTorch 2.1.0 and Python 3.10 as the programming frameworks. The dataset was divided into three subsets: 70% for training, 20% for testing, and 10% for validation. Each sample sequence included 24 time points, enabling the prediction of traffic flow for the next 60 minutes using the data from the previous 60 minutes. Under these configurations, the model underwent extensive training under these configurations for 100 epochs with a learning rate of 0.001. Six attention heads, a 64-dimensional embedding space, and a batch size of 16 were utilized. The Adam optimizer was employed for its efficiency and stability.

D. Experiment Results

TABLE II summarizes the experimental results of the DRSTT model on the PEMS03, PEMS04, and PEMS08 datasets, comparing it with other methodologies. The proposed prediction approach surpasses all baseline models across these datasets and evaluation metrics. In the baseline model, AGCRN with RNN as the main neural network model for sequence prediction showed the best performance among the large class of recurrent neural network models. In comparison to AGCRN, our model achieved improvements in RMSE by 6.33%, 6.76%, and 9.64% on the respective datasets. In the category of spatial-temporal graph convolutional networks, STFGNN demonstrated superior performance by generating data-driven "time graphs" to complement correlations uncaptured by existing spatial graphs. Compared to STFGNN, our model showed improvements in RMSE by 5.47%, 3.39%, and 13.08% on the respective datasets. Among transformer models, Bi-STAT yielded noteworthy results by utilizing dynamic stop modules and bidirectional decoders for prediction. Compared to Bi-STAT, our model achieved RMSE improvements of 2.72%, 0.76%, and 2.81% on the respective datasets. In terms of RMSE for the dataset PEMS03, the DRSTT model in this paper is slightly inferior to the GMAN and GHTPAN models due to the fact that RMSE is more sensitive to large error points and these two models may be better at dealing with the presence of extreme, unevenly distributed

TABLE III
COMPARISON OF 15MIN, 30MIN, 45MIN EXPERIMENTAL RESULTS OF THE OTHER THREE MODELS OF THIS PAPER MODEL

Model	Metric	15min				30min				45min			
		STSGCN	STFGNN	Bi-STAT	DRSTT	STSGCN	STFGNN	Bi-STAT	DRSTT	STSGCN	STFGNN	Bi-STAT	DRSTT
PEMS03	MAE	14.70	14.29	13.63	**12.86**	15.76	15.27	14.29	**13.76**	16.68	16.06	14.86	**14.32**
	RMSE	23.49	24.31	**21.50**	22.26	25.64	26.01	**22.95**	24.19	27.35	27.39	**24.24**	24.80
	MAPE(%)	14.71	14.30	15.09	**13.43**	15.48	15.03	15.08	**14.49**	16.20	15.62	16.05	**14.69**
PEMS04	MAE	18.71	18.24	17.97	**17.03**	19.60	18.86	18.32	**17.55**	20.40	19.37	19.31	**17.95**
	RMSE	29.84	29.33	28.78	**27.92**	31.24	30.35	29.35	**28.81**	32.49	31.18	30.69	**29.47**
	MAPE(%)	12.40	12.13	11.87	**11.32**	12.92	12.46	12.61	**11.66**	13.39	12.75	12.81	**12.18**
PEMS08	MAE	15.14	14.78	14.24	**12.07**	15.79	15.54	14.76	**12.65**	16.41	16.19	15.39	**13.16**
	RMSE	23.29	22.89	22.43	**20.22**	24.46	24.26	23.56	**21.28**	25.51	25.39	24.74	**22.16**
	MAPE(%)	9.89	9.63	9.16	**8.29**	10.21	10.03	9.51	**8.55**	10.57	10.39	9.88	**8.69**

or outliers in the data. However, the better MAE and MAPE of DRSTT may indicate that DRSTT fits the overall traffic pattern more consistently and accurately.The DRSTT model combines multidimensional attributes and combines dynamic routing with converters, and therefore captures the intricate spatial and temporal characteristics of traffic flow well. This innovative approach provides a robust framework for traffic flow prediction with satisfactory demonstration results.

To further demonstrate the efficacy of our model, predictions were conducted across various time horizons—specifically, 15 minutes, 30 minutes, and 45 minutes—and compared with three established baseline models: STSGCN, STFGNN, and Bi-STAT. TABLE III presents a comprehensive comparison of the results obtained by these four models. Notably, in the PEMS04 and PEMS08 datasets, the DRSTT model exhibited superior predictive accuracy across all time intervals of 15, 30, and 45 minutes. Additionally, in the PEMS03 dataset, the DRSTT model achieved the lowest mean absolute error (MAE) and mean absolute percentage error (MAPE), and ranked second in root mean square error (RMSE), demonstrating its strong overall performance. The DRSTT model adeptly captures short-term variations and long-term trends in data, while also accommodating dynamic changes over different time scales. This capability enables accurate predictions across various time horizons. Specifically, for the 15-minute prediction, our model achieved an average MAE of 13.99; for the 30-minute prediction, an average MAE of 14.65; and for the 45-minute prediction, an average MAE of 15.14. These experimental results underscore the high adaptability and precision of the DRSTT model in traffic flow prediction tasks across diverse time steps.

E. Ablation Study

An ablation study was conducted on the PEMS08 dataset to evaluate the effectiveness of each component and step in the proposed DRSTT method. The findings of this study are displayed in TABLE IV. In these experiments, the past 12 time steps were used to predict the following 12 time steps. The analysis concentrated on six key components of the DRSTT model, evaluating the impact of excluding each component:

TABLE IV
ABLATION EXPERIMENT DATA

Architecture	MAE	RMSE	MAPE(%)
DRSTT	**13.48**	**22.79**	**9.03**
W/o PositionEmbedding	13.51	22.96	9.42
W/o SARouting	13.61	22.86	9.14
W/o DynRSST	13.50	22.94	9.03
W/o DynRDST	13.49	22.91	9.17
W/o DynRLTT	13.48	23.02	9.11
W/o DynRSTT	13.60	23.08	9.08

- W/o PositionEmbedding: This variant excludes the position encoding mechanism designed to capture short-term, long-term, and spatial characteristics. Instead, it relies solely on the inherent time series structure and the basic time embedding provided by the data. As shown in TABLE IV, the inclusion of the position embedding module significantly reduces errors across all evaluation metrics, highlighting its role in enhancing positional indexing and improving prediction accuracy.
- W/o SARouting: In this variant, the self-attention routing mechanism is removed. The model uses a traditional transformer for feature extraction instead of the dynamic SARouting module tailored for feature capture. According to TABLE IV, the absence of the SARouting mechanism leads to inferior performance compared to the full DRSTT model. This is due to the dynamic routing's flexibility, which allows the model to adapt information paths dynamically and capture temporal and spatial complexities with greater precision.
- W/o DynRSST module: A variant of this model omits the static part of the spatial routing transformer and relies only on existing methods to capture dynamic spatial and temporal properties. Since traffic models are strongly influenced by static factors such as topology and road capacity, it is important to account for these geographic differences. As can be seen from the TABLE IV, the exclusion of this cell reduces the predictive power, thus emphasizing its importance in the model.
- W/o DynRDST module: This option excludes the dynamic spatial routing transformer module, reducing the model's capacity to detect real-time spatial and temporal features. Considering the critical role of real-time vehicle location and congestion propagation in traffic flow prediction, TABLE IV demonstrates that removing this module leads to a significant overall performance decline compared to the full DRSTT model.
- W/o DynRLTT module: By excluding long-term spatial-temporal routing transformer, the model focuses on capturing only short-term spatial-temporal resources. Given that traffic data is significantly seasonal and cyclical, the inclusion of long-term features is crucial for

TABLE V
SOFT AND HARD ROUTING EXPERIMENT DATA

Model	STT		LTT		SST		DST		MAE	RMSE	MAPE(%)
	hard	soft	hard	soft	hard	soft	hard	soft			
model 1	✓		✓		✓		✓		13.52	22.96	9.04
model 2	✓		✓		✓			✓	**13.48**	**22.79**	**9.03**
model 3	✓		✓			✓	✓		13.51	22.94	9.12
model 4	✓			✓	✓		✓		13.57	22.98	9.14
model 5		✓	✓		✓		✓		13.53	23.02	9.15
model 6	✓		✓			✓		✓	13.50	22.89	9.20
model 7	✓			✓		✓	✓		13.49	22.83	9.03
model 8	✓			✓	✓			✓	13.53	22.98	9.45
model 9		✓		✓	✓		✓		13.50	22.86	9.07
model 10		✓	✓			✓	✓		13.48	22.84	9.11
model 11		✓	✓		✓			✓	13.50	22.97	9.30
model 12	✓			✓		✓		✓	13.46	22.91	9.37
model 13		✓	✓			✓		✓	13.50	23.34	9.10
model 14		✓		✓	✓			✓	13.55	22.87	9.16
model 15		✓		✓		✓	✓		13.59	23.02	9.10
model 16		✓		✓		✓		✓	13.51	22.94	9.08

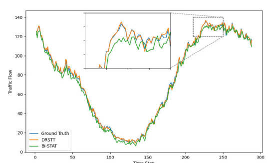

Fig. 4. Visualization for prediction on the PEMS03 dataset. DRSTT is more accurate in predicting details.

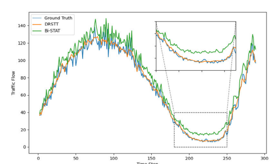

Fig. 5. Visualization for prediction on the PEMS04 dataset. DRSTT is more accurate in predicting details.

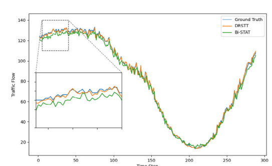

Fig. 6. Visualization for prediction on the PEMS08 dataset. DRSTT is more accurate in predicting details.

accurate prediction. As shown in TABLE IV, omitting the DynRLTT module significantly reduces performance.

- W/o DynRSTT module: This variant lacks the short-term temporal routing transformer, restricting the model to capturing only long-term temporal and spatial features. Since short-term correlations frequently occur within traffic flow across contiguous road segments or nearby regions, TABLE IV shows that models without the DynRSTT module perform worse than the full DRSTT model, indicating the importance of capturing short-term temporal dynamics.

Ablation studies have amply demonstrated the important role of each component in improving the predictive precision and robustness of DRSTT models.

F. Hard and Soft Routing Analysis

The SARouting module adeptly adjusts the temperature parameter (τ) within the Gumbel Softmax framework to facilitate both soft and hard routing mechanisms. This adaptability allows the module to meet the diverse prediction requirements associated with various temporal scales and spatial characteristics. Specifically, when the temperature parameter tau is set to a relatively high value, the Gumbel Softmax yields a more evenly distributed probability distribution, characteristic of "soft routing." In this mode, the model considers multiple potential traffic flow states with a balanced probability distribution, avoiding a preference for any specific output. Conversely, as the temperature parameter tau approaches zero, the probability distribution becomes increasingly concentrated, exhibiting "hard routing" characteristics. Here, the model distinctly selects the most probable traffic flow prediction outcome, rather than distributing probabilities across multiple potential states. To examine the suitability of soft versus hard routing in capturing various temporal and spatial features, rigorous testing was conducted on four distinct modules, utilizing both routing mechanisms independently. Sixteen unique models were designed and comprehensively tested on the PEMS08 dataset. The detailed outcomes of these tests are summarized in TABLE V. Based on these findings, model 2 was identified as the best choice. This preference is mainly due to hard routing's capability to deliver stable and reliable forecasts for both short-term and long-term temporal features, especially those with regular and cyclic patterns. Hard routing is also highly effective in capturing static and fixed-structure features, such as the intrinsic characteristics of the road network. For long-term temporal features, hard routing ensures effective transmission of information across fixed time steps, allowing the model to capture long-term trends and periodic variations, such as seasonal patterns or economic cycles that significantly influence traffic flow. For short-term temporal features, hard routing facilitates deterministic information transmission, accurately simulating transient changes in traffic flow over shorter periods. Additionally, for static spatial features, hard routing leverages consistent paths for information transmission, mirroring the inherent structure of the traffic network and reflecting long-term stable traffic patterns shaped by urban layouts. Dynamic spatial features, however, pose challenges because of their inherent variability and unpredictability. In such contexts, soft routing

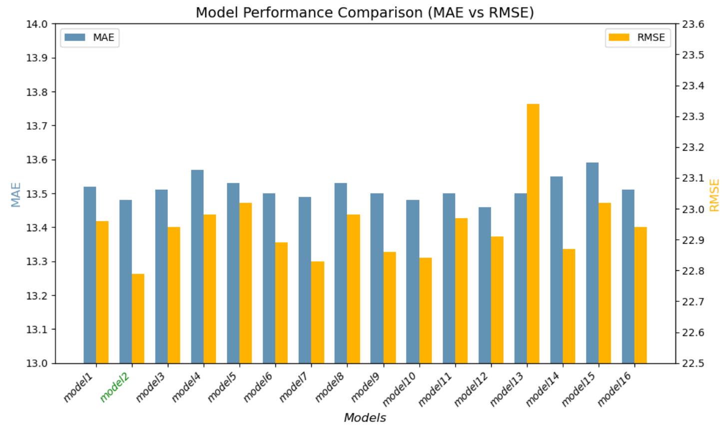

Fig. 7. Visualization of soft and hard routing selection.

demonstrates its advantage by dynamically adjusting the information propagation path based on real-time relationships between nodes, such as fluctuating traffic conditions. Our rigorous experimental evaluation confirms that the DRSTT model effectively selects the most appropriate routing mechanism for different feature types, ultimately enhancing prediction performance.

G. Visual Analysis

This section presents a series of visualizations to offer a deeper insight into the underlying mechanisms of the DRSTT model.

1) Visualization of Prediction Results on Different Datasets: The prediction outcomes of the DRSTT model are compared with those of the Bi-STAT model against the actual traffic flow data from the PEMS03, PEMS04, and PEMS08 datasets, as shown in Fig. 4, Fig. 5, and Fig. 6. These visualizations clearly demonstrate that across various time intervals, the prediction curves generated by the Bi-STAT model often resemble those produced by the DRSTT model. However, as highlighted by the dashed boxes, the DRSTT model demonstrates superior predictive accuracy, especially in complex and challenging scenarios.

2) Performance Visualization of Soft and Hard Routing in Different Feature Modules: Fig. 7 visualizes the performance of different feature modules employing soft and hard routing techniques. The line charts display the MAE and RMSE values derived from the experimental data. Analyzing these charts, it is evident that model 2 exhibits the best overall performance. Moreover, these visualizations align closely with the quantitative data presented in TABLE V.

H. Reliability and Stability Analysis

Fig. 8, Fig. 9, and Fig. 10 show the loss curves of the model on three datasets. The training and validation losses exhibit similar trends across all datasets.

In the early training phase (first 10 epochs), the training loss rapidly decreases from approximately 0.11-0.14 to 0.08-0.09, indicating quick adaptation to the training data. Similarly, the validation loss drops from around 0.12-0.14 to 0.09-0.08 within 20 epochs and stabilizes in the later stages, reflecting good convergence.

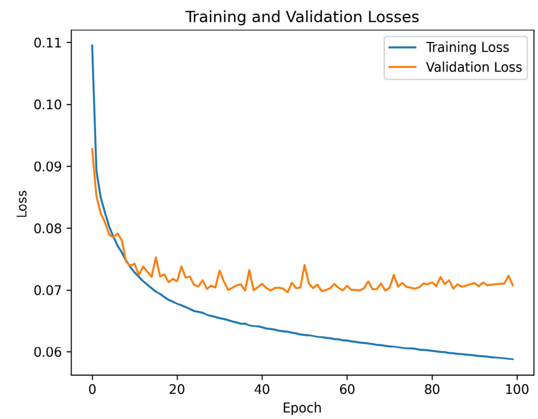

Fig. 8. The loss curve of PEMS03.

Fig. 9. The loss curve of PEMS04.

Minor fluctuations in the validation loss (around 0.005) suggest robustness and no significant instability. The consistent decline and alignment of training and validation losses indicate a balanced fit without overfitting or underfitting. The loss trends are consistent across datasets, with low final validation losses, supporting the model's stability and reliability.

I. Parameters and Computation Time

To thoroughly evaluate the effectiveness of the model proposed in this paper, we conducted experiments focusing on the number of layers in the Dynamic Routing Spatial Temporal

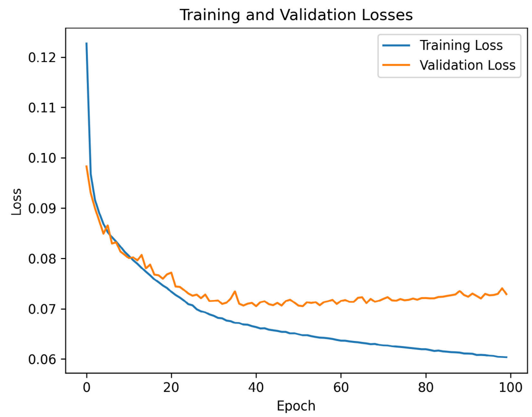

Fig. 10. The loss curve of PEMS08.

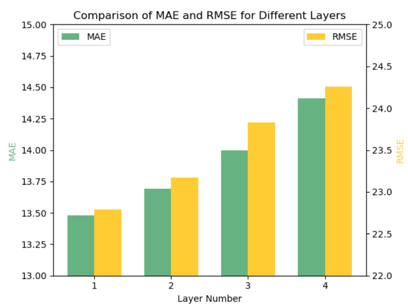

Fig. 11. Histogram of mae,rmse values for different number of layers on dataset PEMS08.

Fig. 12. Histograms of training time and inference time on dataset PEMS08 for different number of layers.

Transformer (DRSTT) module. Specifically, we compared one-layer, two-layer, three-layer, and four-layer configurations using the PEMS08 dataset. The experimental results, including MAE, RMSE, training time, and inference time, are presented in Fig. 11 and Fig. 12.

Fig. 11 illustrates the variation of MAE and RMSE values across different DRSTT module configurations. The results indicate that when the DRSTT module consists of a single layer, the MAE and RMSE values are minimized, demonstrating that this configuration achieves the best predictive performance and highest accuracy. Fig. 12 presents the training and inference times for each configuration. It is evident that with a single-layer DRSTT module, the training time is 52 seconds and the inference time is 5 seconds, both of which are the shortest among all configurations, highlighting the superior efficiency of this configuration. In summary, the experimental results on the PEMS08 dataset confirm that the single-layer DRSTT module excels in both predictive accuracy and computational efficiency, validating its effectiveness as a highly efficient and precise model.

V. Conclusion

This research presents an innovative neural network framework, DRSTT, meticulously designed for traffic flow prediction. This framework is equipped with diverse modules to capture the multifaceted characteristics of spatial-temporal data. By integrating multiple features, DRSTT adeptly addresses the inherent complexities associated with traffic flow. The framework leverages dynamic routing techniques in conjunction with transformers, enabling the network to precisely identify and exploit the intricate interrelationships and interactions among dynamic attributes in traffic flow prediction tasks. This method greatly improves the precision and dependability of the predictions. To deepen the understanding of positional relationships among various elements, the model incorporates distinct positional encodings tailored for each module. The developed model has been rigorously evaluated on three notable real-world traffic flow datasets, confirming its robustness. Both experimental results and theoretical analysis consistently show that DRSTT outperforms current leading methodologies in predictive performance. Although this study introduces innovations in dynamic routing and spatial temporal feature extraction, the model still has limitations in terms of the factors considered. For instance, the focus of this paper has been on offline tasks, and real-time traffic prediction has not been fully explored. Future work could further investigate how to apply the model to real-time prediction scenarios and enhance its adaptability when handling large-scale dynamic data.

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